

HapReel: A Racket-shaped Haptic Display Controller for Presenting Vibrotactile and Force Feedback through Fingertip Deformation

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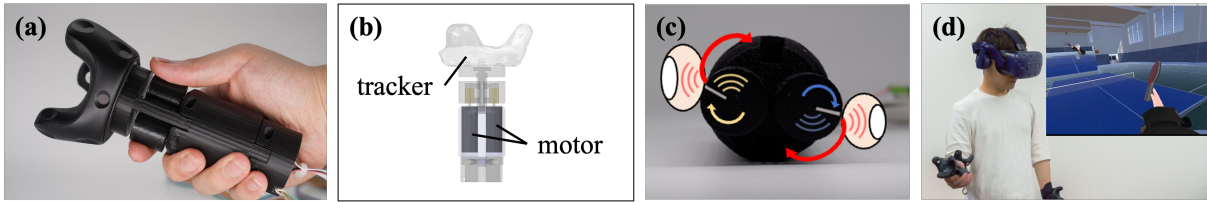


Figure 1: (a) The device is in the shape of a racket and provides tactile feedback. (b) The device is composed of two small motors and a rotor. (c) The rotation and vibration of the motors precisely recreate tactile sensations such as torque and impact. (d) This system makes more immersive VR racket sports possible with a compact device.

ABSTRACT

Racket sports in virtual spaces are rapidly gaining popularity. However, to approximate the real-world experience, particularly when hitting the ball, tactile feedback is crucial. Current feedback is primarily through vibrations and audio, and providing rich tactile feedback typically requires large, heavy devices. This research proposes a method to realize more realistic tactile feedback by presenting force sensations through fingertip deformation using a small device. This novel device has the potential to enable immersive experiences in VR racket sports and aid in the training for racket sports.

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1 INTRODUCTION

The proliferation of affordable VR equipment has led to the widespread adoption of virtual environments for racket sports such as table tennis and tennis. While VR sports offer significant entertainment value, they also contribute to health enhancement and physical conditioning. There have been reported applications in training drills [Liu et al. 2020]. One of the challenges in VR racket sports is the insufficient haptic feedback during ball strikes. Currently, player feedback is predominantly limited to simple vibrations and auditory cues, as seen in commercial game controllers. Haptic feedback plays a critical role in the action of striking a ball in racket sports [Bennett and Beak 2002; Marchal-Crespo et al. 2013]. To provide a richer haptic experience, one often requires devices that are larger, heavier, and more complex than conventional rackets, such as the AirRacket [Tsai et al. 2022].

To address these issues, an approach has been proposed that leverages human perceptual characteristics to provide haptic feedback using smaller devices. For instance, research suggests that the perception of weight can be achieved through fingertip deformation, eliminating the need for actual physical force. Based on this founding, a compact and lightweight device called "Gravity Grabber" has been developed [Minamizawa et al. 2007]. This device uses two motors and belts to modulate the perceived weight sensation of objects being held. Another method has been introduced which presents both force and vibratory stimuli by deforming the fingertip's skin using a DC motor [Nakamura et al. 2017]. A belt, in contact with the skin, conveys haptic feedback. Vibrations are induced by AC signals sent to the motors, while the sensation of force

arises from the skin's deformation caused by DC signals that wind or unwind the belt. This approach enables the delivery of both vibrational and force feedback using a simplified device. In this study, we focused on achieving more immersive haptic feedback in VR racket sports by emphasizing skin deformation, drawing upon insights from previous research. We developed a streamlined haptic feedback display and its associated control methodology. Our goal is to amplify haptic experiences in VR racket sports using a compact and lightweight device that capitalizes on skin deformation. The device, powered by two DC motors, modulates skin deformation on the gripping fingers, delivering both translational and rotational force feedback, as well as vibrational sensations (Fig 1(c)).

2 HAPREAL

The device is composed of a gripping component and a control circuit. The gripping component primarily includes contactors, a casing, two DC motors equipped with encoders, and a motion tracker (Fig 1(b)). Meanwhile, the control circuit consists of a microcontroller, an encoder reading IC, motor drivers, and a power supply. Haptic feedback is facilitated when two fingers (usually the thumb and index finger) come into contact with contactors attached to the DC motors' shafts (Fig 1(a)). The racket component's movement is monitored by the motion tracker, which then mirrors this movement in the virtual environment, ensuring haptic feedback is administered at the correct moments.

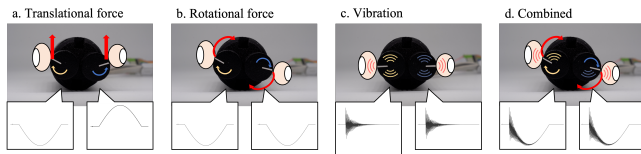


Figure 2: Method of presenting tactile

The device's control methodology is crafted to deliver specific haptic stimuli. In its operation, we devised distinct waveforms for both translational and rotational force feedback, as well as for vibrational feedback. These were then merged into a singular composite waveform. When activating each motor, we employed position control of the DC motor, using the formulated waveforms as target benchmarks. The design of the signal is detailed below.

First, we outline the waveform design intended for delivering translational and rotational force feedback. This device's force feedback aims to simulate the haptic sensation experienced upon the ball's contact with the racket. This sensation is conveyed by twisting the skin of the finger in touch with the device's contactors. Consequently, the created waveform determines the rotation angle of the DC motor. The motor's rotation position, $T(t)$, at time 't' from the onset of the stimulus presentation, is described by the subsequent model equation:

$$T(t) = M \sin(2\pi f t) \quad (1)$$

Here, "M" is the amount of movement from the motor's initial position (the strength of translational or rotational motion), and "f" is a variable determined based on the duration of one stimulus presentation. Using this model equation, translational force feedback

is achieved by rotating the two DC motors in opposite directions. In contrast, rotational force feedback is provided by spinning them in the same direction. Next, we describe the waveform design for providing vibration feedback. The vibrations simulated by this device are primarily based on the vibrations that occur when the ball and racket collide. Therefore, we used a model of a damped sinusoidal wave developed in previous research for waveform generation. The motor's rotation position $D(t)$ at time "t" after the collision of objects is represented by the following model equation:

$$D(t) = A \exp(-Bt) \sin(2\pi f t) \quad (2)$$

Here, "A" is the initial amplitude determined by the collision velocity when objects make contact, "B" is the damping rate of the sinusoidal wave, and "f" is the frequency determined by the material of the colliding objects. As a specific example, we created target waveforms based on this model equation for the case of a ping-pong ball collision (Fig 2(c)). This waveform is a composite of attenuated sinusoidal waves derived from previous research on the vibration spectrum and peak frequency intensity when a ping-pong ball collides with a table tennis racket [Russell 2018].

Finally, the ultimate target rotation position $R(t)$ of the motor is generated by integrating the force feedback waveform $T(t)$ and the vibration waveform $D(t)$ ($R(t) = T(t) + D(t)$). Through these two points of skin deformation feedback, the device enables the presentation of force and vibration sensations using a simplified device (Fig 2(d)). In VR racket sports, this device allows users to not only feel the impact of the ball hitting the racket through a single vibration but also sense the twisting, force, and material characteristics of the racket (Fig 1(d)).

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